

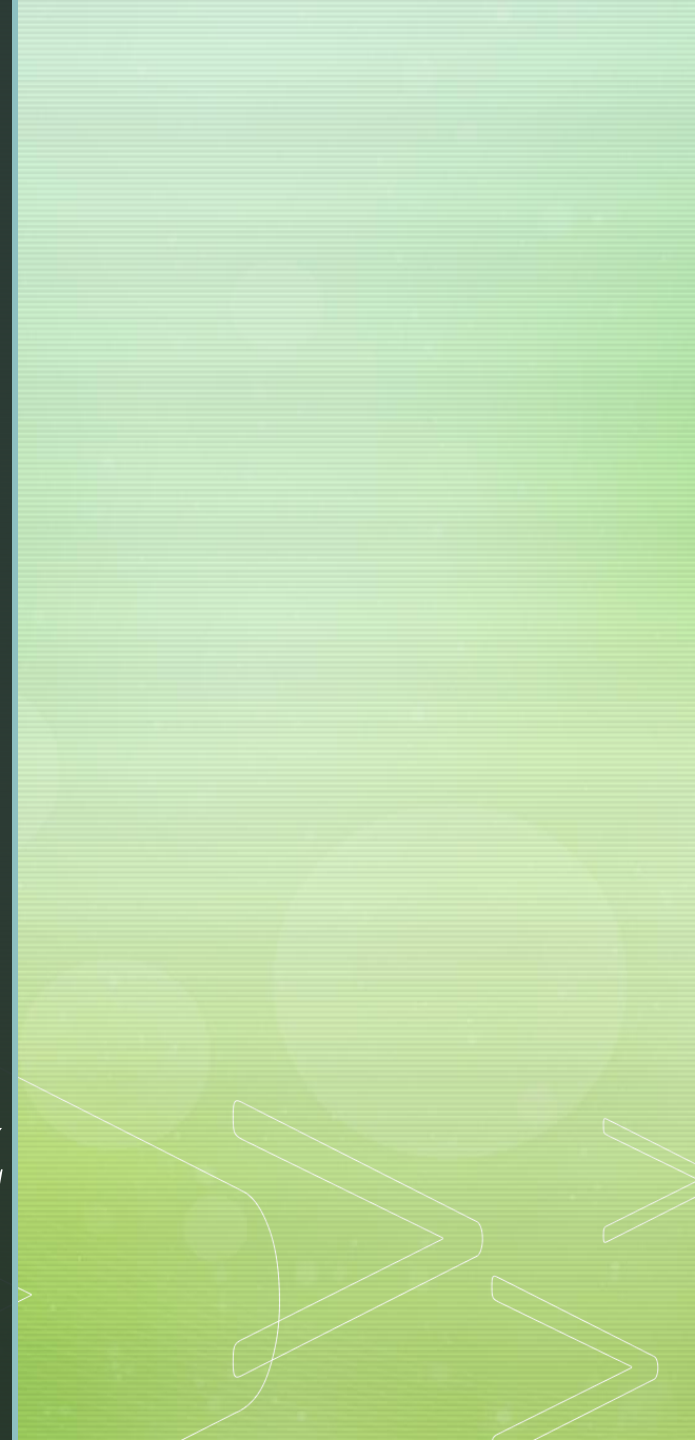
DATABASE MANAGEMENT SYSTEM

CSA0503

TECHNICAL PRESENTATION



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What is a DBMS?

A **Database Management System (DBMS)** is software that stores, manages, and retrieves data.

- It helps users:
 1. Store data safely
 2. Find data quickly
 3. Update data easily
 4. Delete unwanted data
 5. Keep data secure
- **Example:** Think of a **college database**.

It stores:

1. Student details
2. Teacher details
3. Attendance



SOFTWARE COMPONENTS OF DBMS:

A DBMS has **three main software components**:

1.Query Processor

2.Storage Manager

3.Transaction Manager

These components work together to manage data efficiently.

User



Query Processor



Storage Manager



Database



Transaction Manager

- **What is a Query Processor?**

1. The **Query Processor** understands the user's request.

2. A **query** means a request to the database.

- Example:

```
SELECT * FROM Student;
```

- The Query Processor:

1. Reads the query

2. Checks for mistakes

3. Finds the fastest way to get data

4. Sends the request to the Storage Manager

- **Simple Example**

Student asks:

"Show all students."

- The Query Processor understands this request and starts searching.

FUNCTIONS OF QUERY PROCESSOR

- The Query Processor performs four main jobs:

1. Reads the Query

Understands what the user wants.

2. Checks Errors

Makes sure the SQL query is correct.

3. Optimizes the Query

Chooses the fastest method to find data.

4. Sends Request

Passes the request to the Storage Manager.

What is a Storage Manager?

- The **Storage Manager** stores and retrieves data from the database.
- It works like a **librarian**.
- A librarian:
 1. Keeps books safely
 2. Finds books quickly
 3. Returns books to shelves
 4. The Storage Manager does the same for data.
- **Responsibilities**
 1. Stores data
 2. Retrieves data
 3. Updates data
 4. Deletes data
 5. Organizes files

FUNCTIONS OF STORAGE MANAGER

- The Storage Manager performs:

1.Data Storage

Keeps all records safely.

2.Data Retrieval

Finds requested data quickly.

3.Data Update

Changes old information.

Example:

Old Phone Number

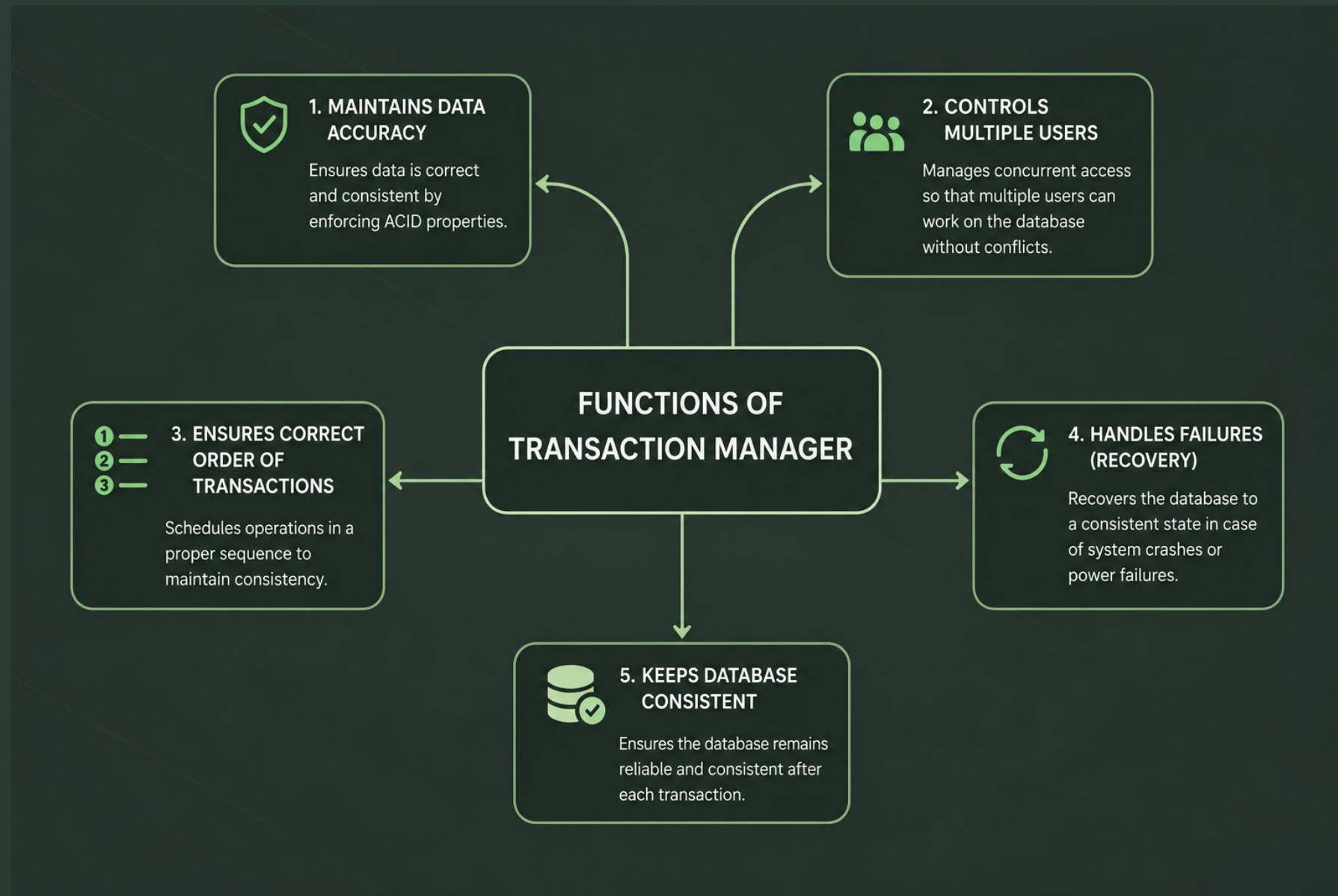


New Phone Number

4.Data Deletion

Removes unwanted records.

- The Storage Manager saves the new attendance in the database



Conclusion

- The **Query Processor**, **Storage Manager**, and **Transaction Manager** work together to make a DBMS **fast, safe, accurate, and reliable**. Without these components, managing large amounts of data would be difficult and error-prone.